1	$2\left(2\cos^2 x - 1\right) = 7\cos x$ $4\cos^2 x - 7\cos x - 2 = 0 \Rightarrow \cos x = -\frac{1}{4}$ $\Rightarrow x = \arccos\left(-\frac{1}{4}\right) = 104.5°, 255.5°$	M1 M1 A1 dM1 A1 **(5)** **(5 marks)**

2.(a)	Sight of $10^{1.478}$ or $10^{0.0646}$ or $10^{0.0646t+1.478}$	M1
	$(a=)$awrt 30 or $(b=)$awrt 1.16	A1
	$\log_{10} N = 0.0646t + 1.478 \rightarrow N = 10^{0.0646t+1.478} = 10^{0.0646t}10^{1.478}$	dM1
	$= "30" \times "1.16"^{t}$	
	$N = 30 \times 1.16^{t}$	A1
		(4)
(b)	Attempts $N = 30 \times 1.16^{30} =$ awrt 2600	M1 A1
		(2)
		(6 marks)

3. (a)	$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{(4x-1)^{\frac{1}{2}}\times 2-(2x+3)\times 2(4x-1)^{-\frac{1}{2}}}{(4x-1)}$	M1 A1
	$\dfrac{(4x-1)^{\frac{1}{2}}\times 2-(2x+3)\times 2(4x-1)^{-\frac{1}{2}}}{(4x-1)}\times\dfrac{(4x-1)^{\frac{1}{2}}}{(4x-1)^{\frac{1}{2}}}=\dfrac{4x-8}{(4x-1)^{\frac{3}{2}}}$	dM1 A1 **(4)**
(b)	Turning point where $\dfrac{\mathrm{d}y}{\mathrm{d}x}=0\Rightarrow x=2$	M1
	Find value of f at $x=2\Rightarrow \mathrm{f}(x)=\sqrt{7}$ Hence range is $\mathrm{f}\geqslant\sqrt{7}$	dM1 A1 **(3)**
		(7 marks)

4 (a)	$\mathrm{ff}(6) = \mathrm{f}(13) = -1$	M1 A1 **(2)**
(b)	Attempts $21+2(2-x)=5x \Rightarrow x=\ldots$ or $21-2(x-2)=5x \Rightarrow x=\ldots$ $x=\frac{25}{7}$ only	M1 A1 **(2)**
(c)	Either $k<21$ or $k \geqslant 17$ $17 \leqslant k < 21$	M1 A1 **(2)**
(d)	$a=\frac{1}{7}$ $b=4$	B1 B1 **(2)** **(8 marks)**

5 (a)	$\sin 3x \equiv \sin(2x+x) \equiv \sin 2x \cos x + \cos 2x \sin x$	M1
	$\equiv 2\sin x \cos x \cos x + \left(1-2\sin^2 x\right)\sin x$	M1
	$\equiv 2\sin x\left(1-\sin^2 x\right)+\left(1-2\sin^2 x\right)\sin x$	ddM1
	$\equiv 3\sin x - 4\sin^3 x$ *	A1*
		(4)
(b)	$\int_0^{\frac{\pi}{3}} \sin^3 x \, \mathrm{d}x = \int_0^{\frac{\pi}{3}} \frac{3}{4}\sin x - \frac{1}{4}\sin 3x \, \mathrm{d}x$	M1
	$= \left[-\frac{3}{4}\cos x + \frac{1}{12}\cos 3x\right]_0^{\frac{\pi}{3}}$	dM1 A1
	$= \frac{5}{24}$	A1
		(4)
		(8 marks)

5 (b) Alt 1

$$\int_0^{\frac{\pi}{3}} \sin^3 x \, dx = \int_0^{\frac{\pi}{3}} \sin x\left(1-\cos^2 x\right) dx = \int_0^{\frac{\pi}{3}} \sin x - \sin x \cos^2 x \, dx$$

M1

$$= \left[-\cos x - \left(-\frac{\cos^3 x}{3}\right)\right]_0^{\frac{\pi}{3}}$$

dM1A1

$$= \frac{5}{24}$$

A1

(4)

(b) Alt 2

$$u = \cos x \Rightarrow \frac{du}{dx} = -\sin x \Rightarrow \int_0^{\frac{\pi}{3}} \sin^3 x \, dx = \int_1^{\frac{1}{2}} u^2 - 1 \, du$$

M1

$$= \left[\frac{u^3}{3} - u\right]_1^{\frac{1}{2}}$$

dM1 A1

$$= \frac{5}{24}$$

A1

(4)

(8 marks)

6 (a)	$5e^{x-1}+3=18 \Rightarrow e^{x-1}=3$	M1
	$\Rightarrow x=\ln 3+1$ or $e^{x}=3e$	A1
	$\Rightarrow x=\ln 3e$	A1
		(3)
(b)	Sets $5e^{x-1}+3=10-x^{2}$ and proceeds to find and use a suitable function. Eg $\left(f(x)=\right)7-x^{2}-5e^{x-1}$	B1
	Attempts $f(1.1335)=0.001$ and $f(1.1345)=-0.007$	M1
	Correct values with reason(change of sign and continuous) and conclusion, hence α is 1.134 to 3dp	A1
		(3)
(c)	$x_2=-\sqrt{7-5e^{-3-1}}=-2.628388$	M1 A1
	$\beta=-2.620330$	A1
		(3)
		(9 marks)

7.(a)	$R = \sqrt{17}$ $\tan\alpha = 4 \Rightarrow \alpha =$ awrt 1.326	B1 M1A1 **(3)**
(b)	Minimum height $= \dfrac{24}{3+"R"} = 3.37$ (metres)	M1 A1 **(2)**
(c)	Uses part (a) $10 = \dfrac{24}{3+\sqrt{17}\cos\left(\frac{1}{2}t - 1.326\right)} \Rightarrow \cos\left(\frac{1}{2}t - 1.326\right) = \dfrac{-0.6}{\sqrt{17}}$ $t =$ awrt 6.09	M1 A1 M1 A1 **(4)** **(9 marks)**

8(i)(a)	$g'(x) = 3e^{3x}\sec 2x + 2e^{3x}\sec 2x\tan 2x$	M1 A1 **(2)**
(b)	$g'(x) = 0 \Rightarrow e^{3x}\sec 2x(3 + 2\tan 2x) = 0$	M1
	$\tan 2x = -1.5 \Rightarrow x = -0.491$	dM1 A1 **(3)**
(ii)	$x = \ln(\sin y) \Rightarrow \dfrac{dx}{dy} = \dfrac{1}{\sin y} \times \cos y$	B1
	Attempts $\cos y = \sqrt{1-\sin^2 y} = \sqrt{1-e^{2x}}$ or $\dfrac{dy}{dx} = \dfrac{\sin y}{\cos y}$	M1
	Hence $\dfrac{dy}{dx} = \dfrac{\sin y}{\cos y} = \dfrac{e^x}{\sqrt{1-e^{2x}}}$	dM1 A1 **(4)** **(9 marks)**

9 (a)	$$\begin{array}{r} x^2+2 \\ x^2-x-12\overline{)x^4-x^3-10x^2+3x-9} \\ \underline{x^4-x^3-12x^2} \\ 2x^2+3x-9 \\ \underline{2x^2-2x-24} \\ 5x+15 \end{array}$$	M1A1
	$\dfrac{x^4-x^3-10x^2+3x-9}{x^2+x-12} \equiv x^2+2+\dfrac{5(x+3)}{(x-4)(x+3)}$	M1
	$\equiv x^2+2+\dfrac{5}{(x-4)}$ or stating $P=2, Q=5$	A1
		(4)
(b)	$g'(x)=2x-\dfrac{5}{(x-4)^2}$	M1A1
	Subs $x=2$ into $g'(2)=2\times 2-\dfrac{5}{(2-4)^2}=\dfrac{11}{4}$	M1
	Uses $m=g'(2)=\left(\dfrac{11}{4}\right)$ with $(2, g(2))=\left(2,\dfrac{7}{2}\right)$ to form equation of tangent	
	$y-\dfrac{7}{2}=\dfrac{11}{4}(x-2) \Rightarrow y=\dfrac{11}{4}x-2$	dM1A1
		(5)
(c)	$\int x^2+2+\dfrac{5}{(x-4)}\,dx=\dfrac{1}{3}x^3+2x,+5\ln\lvert x-4\rvert$	M1 A1ft
	Area $R=\left[\dfrac{1}{3}x^3+2x+5\ln\lvert x-4\rvert\right]_0^2=\left(\dfrac{8}{3}+4+5\ln 2\right)-\left(0+0+5\ln 4\right)$	dM1
	$=\dfrac{20}{3}+5\ln 2-5\ln 4=\dfrac{20}{3}-5\ln 2$	ddM1 A1
		(5)
		(14 marks)
Alt (a)	$x^4-x^3-10x^2+3x-9 \equiv (x^2+P)(x^2-x-12)+Q(x+3)$	M1
	Compare terms (OR sub in values) and solve simultaneously to find P and/or Q ie $x^2 \Rightarrow P-12=-10$, $x \Rightarrow -P+Q=3$, const $\Rightarrow -12P+3Q=-9$ $\Rightarrow P=\ldots$ or $Q=\ldots$	M1
	$P=2, Q=5$ *	A1,A1
	Award in the order shown here on ePEN.	